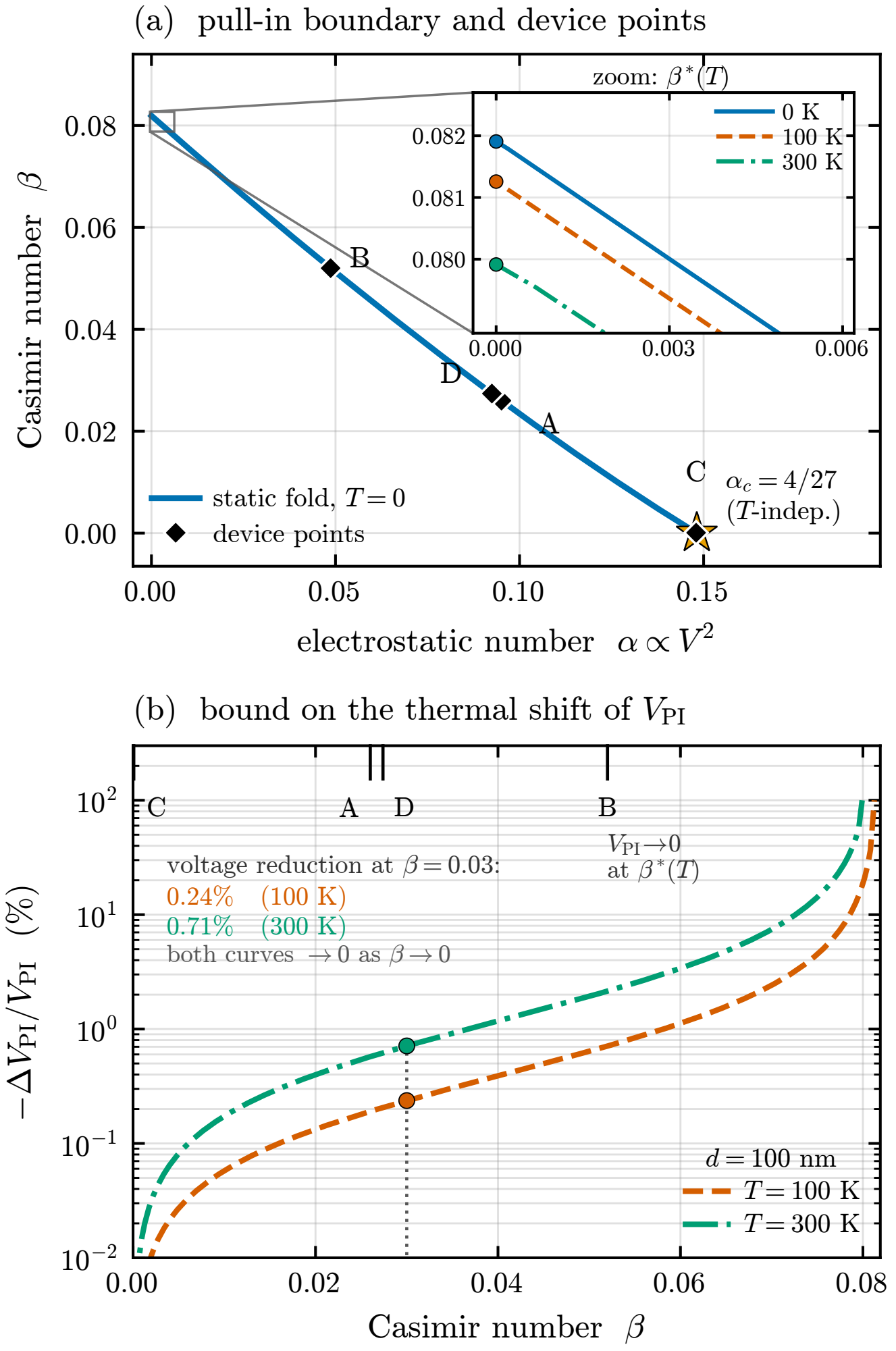
(a) pull-in boundary and device points
zoom: $\beta^*(T)$
0 K
100 K
300 K
0.082
0.081
0.080
0.000
0.003
0.006
B
D
A
C
$\alpha_c = 4/27$
(T-indep.)
static fold, $T = 0$
device points
Casimir number β
0.08
0.06
0.04
0.02
0.00
0.00
0.05
0.10
0.15
electrostatic number $\alpha \propto V^2$
(b) bound on the thermal shift of V_{PI}
C
A
D
B
voltage reduction at $\beta = 0.03$:
0.24% (100 K)
0.71% (300 K)
both curves $\to 0$ as $\beta \to 0$
$V_{\mathrm{PI}} \to 0$
at $\beta^*(T)$
$d = 100$ nm
$T = 100$ K
$T = 300$ K
$-\Delta V_{\mathrm{PI}}/V_{\mathrm{PI}}$ (%)
10^2
10^1
10^0
10^{-1}
10^{-2}
0.00
0.02
0.04
0.06
0.08
Casimir number β